

**MINISTRY OF EDUCATION AND RESEARCH OF THE REPUBLIC OF
MOLDOVA**

TECHNICAL UNIVERSITY OF MOLDOVA

Faculty of Computers, Informatics and Microelectronics
Department of Software Engineering and Automation

LABORATORY WORK No. 2

Cryptanalysis of Monoalphabetic Ciphers

Variant V11

Performed by:

Student, group FAF-243
Andrei Elciscev

Checked by:

Zaica Maia

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1 Objective

The objective of this laboratory work is to study the cryptanalysis of monoalphabetic ciphers using letter frequency analysis and to apply this method to decrypt an intercepted message written in English.

The specific objectives are:

- to understand how monoalphabetic substitution ciphers work;
- to analyze the frequency of letters in an encrypted text;
- to identify possible correspondences between ciphertext and plaintext letters;
- to use word patterns and context to verify substitutions;
- to reconstruct the original message for Variant V11.

2 Theoretical Background

2.1 Letter Frequency Analysis

A monoalphabetic substitution cipher is an encryption system in which each plaintext letter is replaced with a corresponding ciphertext letter. In a simple monoalphabetic substitution, the same plaintext letter is always replaced by the same ciphertext letter.

The main weakness of this system is that it preserves the relative frequency pattern of letters. In a sufficiently long encrypted text, the frequency distribution of ciphertext letters reflects, to some extent, the distribution of letters in the language of the original message.

In English, the letters E, T, A, O, I, and N occur frequently. Therefore, if a particular letter appears very often in a ciphertext, it may represent one of these common plaintext letters.

However, this assumption is not always correct. Letter frequencies vary between texts depending on their subject, length, and vocabulary. For this reason, frequency analysis should be combined with the examination of word patterns, letter pairs, three-letter combinations, and the overall meaning of the message.

2.2 Frequency Analysis Methodology

A frequency analysis attack can be performed using the following steps:

1. Identify all letters occurring in the ciphertext.
2. Count the occurrences of each letter.
3. Sort the letters in descending order of frequency.
4. Compare the resulting distribution with typical English letter frequencies.

5. Form hypotheses about possible substitutions.
6. Apply the substitutions and examine the resulting words.
7. Verify the hypotheses using common words, grammatical structure, and the context of the message.
8. Repeat the process until the text can be reconstructed and the substitutions are sufficiently verified.

In English, words such as *the*, *and*, *was*, *under*, and *with* can help verify substitution hypotheses. Short words and repeated patterns can also provide useful clues.

2.3 Mean Frequency of Letters in the English Alphabet

Frequency analysis relies on the fact that letters do not occur equally often in English texts. The following table presents the approximate average frequency of each letter in English, expressed as a percentage of all letters in a typical text. These values serve as a reference when analyzing the ciphertext and identifying possible substitutions.

Table 1: Approximate Letter Frequencies in English

Letter	Freq. (%)	Letter	Freq. (%)	Letter	Freq. (%)	Letter	Freq. (%)
A	8.17	H	6.09	O	7.51	U	2.76
B	1.49	I	6.97	P	1.93	V	0.98
C	2.78	J	0.15	Q	0.10	W	2.36
D	4.25	K	0.77	R	5.99	X	0.15
E	12.70	L	4.03	S	6.33	Y	1.97
F	2.23	M	2.41	T	9.06	Z	0.07
G	2.02	N	6.75				

As shown in Table 1, the most frequent letters in English are typically E, T, A, O, I, and N. Conversely, letters such as Q, X, Z, and J occur relatively rarely. When analyzing a monoalphabetic substitution cipher, the most frequent ciphertext symbols can therefore be compared with these common English letters to suggest possible substitutions. However, frequency alone does not prove a substitution; letter patterns, common words, and the consistency of the resulting plaintext must also be considered.

3 Intercepted Ciphertext

Variant V11 consists of a message encrypted using a monoalphabetic substitution cipher. It is known that the original message was written in English and that only letters were encrypted. Spaces, punctuation marks, and other characters remain unchanged.

The ciphertext begins with the following sequence:

IVTSZNGW rtp dgovi pxvjv.

The message contains several paragraphs describing historical events in seventeenth-century France, including the siege of Réalmont and the activities of the cryptanalyst Antoine Rossignol.

The length of the message provides sufficient material for frequency analysis, while the preserved spaces allow the structure of individual words to be examined.

4 Cryptanalysis Process

4.1 Identifying the Initial Words

The first step was to examine the beginning of the ciphertext:

IVTSZNGW rtp dgovi pxvjv.

By examining the word patterns and forming hypotheses based on the English language, the sequence can be interpreted as:

REALMONT was under siege.

This interpretation provides an initial set of correspondences between the ciphertext alphabet and the plaintext alphabet.

4.2 Initial Substitutions

The following table presents the substitutions inferred from the opening words.

Table 2: Initial substitutions identified

Ciphertext Letter	Plaintext Letter	Example
I	R	IVTSZNGW → REALMONT
V	E	IVTSZNGW → REALMONT
T	A	IVTSZNGW → REALMONT
S	L	IVTSZNGW → REALMONT
Z	M	IVTSZNGW → REALMONT
N	O	IVTSZNGW → REALMONT
G	N	IVTSZNGW → REALMONT
W	T	IVTSZNGW → REALMONT
R	W	rtp → was
P	S	rtp → was
D	U	dgovi → under
O	D	dgovi → under
X	I	pxvjv → siege
J	G	pxvjv → siege

These correspondences can be used to test other words in the ciphertext. If a substitution produces meaningless words or contradicts other established correspondences, the hypothesis must be reviewed and, if necessary, corrected.

4.3 Verification Using Context

After the initial substitutions, words and expressions appearing elsewhere in the message can be examined. Repeated patterns help confirm previously identified letters and reveal additional substitutions.

In this case, the historical context provides useful clues. The text refers to the Huguenots, Cardinal Richelieu, the city of La Rochelle, and the cryptanalyst Antoine Rossignol. These references help verify the result, but they do not replace the actual analysis of the ciphertext.

Throughout the process, each ciphertext letter must consistently correspond to the same plaintext letter. This is an essential property of a monoalphabetic substitution cipher.

4.4 Reconstructing the Message

Once enough substitutions have been identified, they can be applied throughout the ciphertext. The resulting words are checked against English grammar and the overall meaning of the passage.

The process can be summarized as follows:

1. identify the most frequent letters;
2. formulate hypotheses about possible substitutions;
3. verify the hypotheses using recognizable words;
4. extend the substitution mapping to additional letters;
5. apply the mapping to the entire ciphertext;
6. verify the coherence of the reconstructed message.

5 Decrypted Message

The following passage represents the reconstructed English text associated with Variant V11.

REALMONT was under siege. The royal army, under Henry II of Bourbon, Prince of Condé, had invested it at dawn Wednesday, April 19, 1628. But the Huguenots, inside the battlements of the little town in southern France, were putting up a stiff defense. They cannonaded Condé from a tower and contemptuously rejected his demands that they surrender, saying that they would die instead. Condé brought up five big cannon from Albi, a dozen miles away, and on Sunday ranged them in an ominous line facing Realmont.

That same day his soldiers captured an inhabitant of the town who was trying to carry an enciphered message to Huguenot forces outside. None of Condé's men could unriddle

it, but during the week the prince learned that it might be solved by the scion of a leading family of Albi who was known to have an interest in ciphers.

Condé sent him the cryptogram. The young man solved it on the spot. It revealed that the Huguenots desperately needed munitions and that, if they were not supplied, they would have to yield. This was news indeed, for despite the destruction of a number of houses by the Catholic batteries, the town was continuing to resist stoutly with no sign of surrender. Condé returned the cryptogram to the inhabitants, and on Sunday, April 30, 1628, though its fortifications were still unbreached and its defenses still apparently adequate for a long siege, Realmont suddenly and unexpectedly capitulated. With this dramatic success began the career of the man who was to become France's first full-time cryptologist: the great Antoine Rossignol.

When word of the incident reached Cardinal Richelieu, the astute and able Gray Eminence of France, he at once attached this useful talent to his suite. Rossignol proved his worth almost immediately. The Catholic armies under Richelieu surrounding the chief Huguenot bastion of La Rochelle intercepted some letters in cipher, which the young codebreaker of Albi read with ease. He told His Eminence that the starving citizens were eagerly awaiting help that the English had promised to send by sea. When the fleet arrived, the primed guardships and forts so intimidated it that it stood off the port's entrance and made no serious attempt to force a passage. A month later, the city capitulated in full sight of the English vessels—and the great French tradition of expertise in cryptology had been founded.

Rossignol very quickly established himself in the royal service. By 1630, his solutions had made him rich enough to build a small but elegant chateau at Juvisy, 12 miles south of Paris, later surrounding it with a charming informal garden designed by Le Notre, the gardener of Versailles. Here Louis XIII stopped to visit the young crypt-analyst in 1634, 1635 and 1636 on his returns to Paris from Fontainebleau.

In the swashbuckling court of that monarch, and then in the resplendent one of Louis XIV, Rossignol served with an extraordinary facility. The stronghold of Hesdin surrendered a week sooner than it otherwise would have because he solved an enciphered plea for help, and then composed a reply in the same cipher telling the townspeople how futile their hopes were. How many other towns he compelled to surrender, how many diplomatic coups he made possible, how many betrayals he uncovered among the great nobles in those days of shifting allegiances, he never discussed.

This reticence caused some at the court to charge that he never actually solved a single cipher, and that the cardinal spread inflated rumors about his abilities to discourage would-be conspirators. But in fact Richelieu was frequently telling his subordinates such things as, "It is necessary to make use, in my opinion, of the letters of the man who has been arrested by the civil authorities at Mezieres, that is to say, have them put into Rossignol's hands to see if there is something important in them." Or, eight years later, in 1642, writing to Messieurs de Noyers and de Chavigny: "I saw, in some extracts, that Rossignol sent me, a truce negotiation of the King of England with the Prince of Orange; I do not think that it can have any effect, but . . . it is up to you, gentlemen, to

keep your eyes peeled.”

6 Conclusions

This laboratory work examined letter frequency analysis and its application to the cryptanalysis of a monoalphabetic substitution cipher.

The cryptanalysis process involves identifying letter frequencies, forming hypotheses about substitutions, and verifying them by examining word patterns and context. Frequency analysis provides useful clues, but it is not sufficient on its own to guarantee the correct decryption of every message.

The exercise demonstrates the vulnerability of monoalphabetic ciphers when sufficiently long messages are available. Since each letter is replaced consistently, statistical and linguistic characteristics of the original text remain visible in the ciphertext.

In conclusion, combining letter frequency analysis with word-pattern examination and contextual reasoning allows the original message to be progressively reconstructed. This illustrates why simple monoalphabetic substitution does not provide sufficient cryptographic security.